INCH-POUND

MIL-DTL-25866C
24 May 2010
SUPERSEDING
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24 October 1969

DETAIL SPECIFICATION

LIGHT, EMERGENCY EXIT, AIRCRAFT LEU-1/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of emergency exit light for cargo-type aircraft, designated LEU-1/A.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-B-18	- Batteries, Non-Rechargeable, Dry
MIL-DTL-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-A-8625	- Anodic Coatings for Aluminum and Aluminum Alloys
MS35059	- Switch, Toggle, Two Pole, Sealed Toggle

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-202	- Electronic and Electrical Component Parts
MIL-STD-810	- Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME INTERNATIONAL

ASME Y14.100	- Engineering Drawing Practices

(Copies of this document are available online at http://www.asme.org/ or from ASME International, Three Park Avenue, New York, NY 10016-5990.)

ASTM INTERNATIONAL

ASTM B117	- Operating Salt Spray (Fog) Apparatus

(Copies of this document are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE AMS-QQ-P-416	- Plating, Cadmium (Electrodeposited)

(Copies of this document are available from http://www.sae.org/ or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article testing in accordance with 4.2.

3.2 Components. The light shall consist of:

Item No.	Quantity	Description	Requirement
1	1	Housing	3.6
2	1	Base plate	3.7
3	1	Reflector	3.8
4	1	Lens	3.9
5	2	Lamp	3.10
6	1	Manual switch	3.11
7	1	Inertia switch	3.12
8	1	Power failure relay	3.13
9	4	Batteries	3.14

3.3 Materials.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4 Design and construction. The light shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strain, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Adjustment and repair. The light shall be so constructed that adjustments and repairs (including replacement of lamp and batteries) can be made easily by the personnel of operating units, with tools normally available commercially.

3.4.2 Installation. The light shall be designed for installation near each exit of the aircraft cabin so that the light beam will be directed to illuminate the exit in an efficient manner. All parts shall be suitably enclosed to prevent tampering or damage.

3.4.3 Portability. The light shall be so designed that it can be easily and quickly removed from its mounting position. The device used to secure the light in its mounting position shall hold the light in place during the conditions specified in 3.5.

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3.4.3.1 Handle. A handle shall be provided for carrying the light. The configuration of the handle to be provided shall be subject to approval by the procuring activity.

3.4.4 Lights of different construction. Lights of slightly different construction will be considered to be in accordance with the requirements, provided the specified and limiting dimensions, mounting holes, connections, and all other requirements specified herein are met and the design is approved by the procuring activity.

3.5 Performance. Unless otherwise specified, the light shall be capable of operating satisfactorily under the following conditions:

a. Temperature ranging from -65 to 160 °F (-54 to 71 °C), except for reduction in-battery capacity.

b. Pressures ranging from 30 to 13.7 inches of mercury (inHg) (approximately an altitude of 20,000 feet).

c. Relative humidity up to 95 percent including conditions wherein condensation takes place in the form of both water and frost.

d. Sand and dust particles as will be encountered in desert areas.

e. Exposure to salt-sea atmosphere.

f. Operation in an explosive vapor within or surrounding the equipment.

g. Vibration incident to service use.

h. Acceleration and shock forces as anticipated.

i. Underwater operation.

3.5.1 Illumination (photometric). The light shall be designed in such a manner as to provide the following illumination. When properly installed and operated at rated lamp voltage, the candlepower distribution in any plane containing an axis which passes through the light center and is perpendicular to the mounting plate (0 degree axis) shall equal or exceed the values in table I.

TABLE I. Candlepower distribution.

Candlepower	Minimum width with respect to 0° axis
7	-5° to +5°
5	-10° to +10°
3	-15° to +15°

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3.6 Housing. The housing shall be designed in such a manner as to support the light and it may utilize as much of the base plate area as required. If the housing obstructs the mounting holes, the light shall be so designed that the mounting holes are readily accessible when the cover of the housing is removed.

3.7 Base plate. The design of the light shall incorporate a plate to provide a supporting base for the housing. The light shall be such that it may be attached directly to the cabin structural members or to a suitable bracket by means of the mounting holes in the base plate. The mounting holes and a hole to accommodate the electrical receptacle mounted on the housing shall be located as shown on figure 1.

3.7.1 Flight axis. The base plate shall be marked to indicate the proper positioning alignment of the light with the flight axis of the aircraft. The light shall be indexed with respect to the base plate.

3.8 Reflector. The light shall be provided with a suitable polished-aluminum reflector.

3.9 Lens. The light shall be provided with a white-plastic diffusing lens.

3.10 Lamp. The light shall be furnished with a single-contact miniature-flanged base, B-3.5 bulb, 2.38 volts, 0.50 ampere, commercial, PR-2 flashlight lamp. The light shall be provided with a spare lamp complete with lamp holder to be mounted conveniently inside the housing. If lamp is of a type requiring continuous recharge of batteries from ship power source, a pilot light shall indicate when power is on and recharging batteries.

3.11 Manual switch. The light shall be provided with an MS35059-21 switch. The positive terminal of the self-contained battery power supply shall be attached to the center (OFF) terminal of this switch. One of the ON terminals shall be connected directly to the lamp. The circuit from the second ON terminal shall pass through the set of normally closed contacts of the power failure relay, and thence to the lamp. When the switch is in this second ON position, the light shall then be armed for automatic operation by the power failure relay. The switch shall be located either in the light housing or external to the housing.

3.12 Inertia switch. The light shall be provided with an inertia switch, which will automatically energize the light when subjected to a deceleration of 2 ±0.5 gravity (G). The direction of maximum sensitivity (minimum impact force required to energize the light) shall be as marked on the base plate. The inertia switch shall be so designed that the light will not come on when subjected to incident vibration as encountered in aircraft. If the inertia switch is of the maintained contact type, it shall incorporate a push button, mechanical reset device so that the light may quickly be rearmed for automatic operation. If the inertia switch is comprised of a momentary contact device in conjunction with a holding relay, a single circuit, normally closed push-type switch may be utilized as a means of breaking the circuit of the holding relay in order to rearm the light for automatic operation. The switch shall be located either in the light housing or external to the housing.

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3.13 Power failure relay. The light shall be provided with an approved relay, which has a set of normally closed contacts. The coil of the relay shall be energized in the armed position from the aircraft electrical power circuits (28 volts direct current (VDC)). The operation of the relay shall be such that whenever the aircraft electrical power fails or is turned off, the contacts will be closed and the lamp will be automatically energized from the self-contained battery power supply.

3.13.1 Rearming circuit. The light shall be so designed that it can be rearmed by the light switch, and remotely rearmed if so connected. Figure 2 illustrates a circuit by which this requirement may be obtained. Other circuits will be acceptable provided the requirements are met. If the light is to be de-energized and rearmed remotely, the control switch shall be connected to terminal C.

3.13.2 Electrical connections. Four 8-32 screw-type terminals shall be provided to connect the power and control circuits (if applicable) to the light. The terminals shall be marked A, B, C, and D. Terminal A shall connect to the +28 VDC circuit and terminal B shall connect to ground (-28 VDC circuit). Terminal C is provided for remote de-energization and rearming of the light if required.

3.13.3 Light turn-on feature. Provisions shall be incorporated so that upon removal of the light from its mounting plate, the light will automatically energize in the ON position. This may be accomplished either physically or electrically.

3.14 Batteries. The light shall be powered by four standard BA-30 cells conforming to MIL-B-18. The batteries shall be arranged in series parallel, so as to provide a nominal open-circuit voltage of 3 volts. The batteries shall be used only to provide power for the lamp and under no conditions shall be used as a power source for any other component part of the light such as a relay or other device. Unless otherwise specified (see 6.2), the batteries shall not be furnished with the light.

3.15 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of ASME Y14.100, numbering, coding, and identification section, shall govern changes in the manufacturer's part numbers.

3.16 Dimensions. The dimensions of the light shall remain within the envelope specified in figure 1.

3.17 Weight. The weight of the light with lamp and batteries installed shall be not more than 3.5 pounds.

3.18 Finishes and protective coatings.

3.18.1 Aluminum-alloy parts. Aluminum-alloy parts shall be covered with an anodic film in accordance with MlL-A-8625, except as follows:

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a. Small holes need not be anodized.

b. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-DTL-5541.

c. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIL-DTL-5541 may be used in lieu of anodizing.

d. Castings containing nonaluminum-alloy integral inserts may be treated with a chemical film in accordance with MIL-DTL-5541.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-DTL-5541 shall not be used in lieu of anodizing.

3.18.2 Steel parts. Steel parts shall be cadmium plated, where practicable, in accordance with SAE AMS-QQ-P-416, type II or III, and of a class that is adequate to achieve the degree of protection required.

3.19 Operating instructions. Instruction plates containing operating instructions such as ON, OFF, RESET, shall be permanently affixed to the light. The lettering shall be clearly legible and permanently marked.

3.20 Wiring diagram. A circuit diagram with a legend describing the electrical components of the light shall be permanently attached to the interior of the housing or other surface where practicable. The diagram shall be legible and protected against destruction by any of the tests specified in section 4. Proper method for inserting the batteries shall be indicated.

3.21 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130, except that serial numbers are not required.

3.22 Workmanship. All details of workmanship and finish shall be in accordance with the best practice for high quality emergency exit lights consistent with the requirements of this specification. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, and bolts.

3.22.1 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.22.2 Cleaning. The light shall be thoroughly cleaned. Excess solder, metal chips, and other foreign material shall be removed during and after final assembly.

3.23 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible,

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provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article testing (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article testing.

4.2.1 First article test samples. The first article test samples shall consist of three lights representative of the production equipment. They shall be tested in a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.2.2 First article tests. First article tests shall consist of all tests described under 4.5 and shall be tested in the order show in table II.

TABLE II. First article test sequence.

Sample number 1	Sample number 2	Sample number 3
Examination of product Illumination (photometric) High temperature Low temperature Altitude	Examination of product Vibration Acceleration Shock Underwater operation Portability	Examination of product Humidity Salt spray Sand and dust Explosion proof

4.3 Conformance inspection. Conformance inspection shall consist of the individual test.

4.3.1 Individual test. Each light shall be subjected to the examination of product described in 4.5.1 and 4.5.1.1.

4.4 Test conditions.

4.4.1 Atmospheric conditions. Unless otherwise specified, the tests shall be made at an atmospheric pressure of approximately 29.92 inHg and at room temperature of approximately 77 °F (25 °C). Where tests are made with pressure or temperature differing materially from these values, proper allowance shall be made for the difference from the specified condition.

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4.4.2 Circuit conditions. The test specimen shall be equipped with a PR-2 flashlight lamp, batteries, and unless otherwise specified, 28 VDC shall be applied to the power-failure relay when the light is subjected to tests which require functional operation of the light.

4.5 Test methods.

4.5.1 Examination of product. The light shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking.

4.5.1.1 Switch operation. The light, with lamp and batteries installed, shall be inspected for functional operation of inertia switch and reset, manual switch, and for the proper operation of the relay or other device that energizes the light when aircraft electrical power is off.

4.5.2 Environmental tests.

4.5.2.1 High temperature. The complete light with batteries installed shall be subjected to high temperature in accordance with MIL-STD-810, method 501, procedure III, at a temperature of 160 ±3.6 °F (71 ±2 °C), dry heat, for a period of 4 hours, followed by an operational test of the manual switch, and proper operation of the relay, or other device, which energizes the light when aircraft electrical power is lost. This test shall be conducted while at the high temperature. At the end of the tests, the light shall be removed from the test chamber and examined. There shall be no crazing, dimensional changes, or other defects. The light shall be returned to room temperature and checked for proper operation.

4.5.2.2 Low temperature. The complete light with batteries installed shall be subjected to low temperature in accordance with MIL-STD-810, method 502, procedure II, at a temperature of -65 ±3.6 °F (-54 ±2 °C), for a period of 4 hours, followed by an operational test of the manual switch, and proper operation of the relay, or other device, which energizes the light when aircraft electrical power is lost. This test shall be conducted while at the low temperature. At the end of the tests, the light shall be removed from the test chamber and examined. There shall be no crazing, dimensional changes, or other defects. The light shall be returned to room temperature and checked for proper operation.

4.5.2.3 Altitude. The assembly shall be subjected to an altitude test in accordance with MIL-STD-810, method 500, procedure II, at an altitude of 20,000 feet, a temperature of 32 °F (0 °C), for a period of 4 hours, followed by an operational test of the manual switch, and proper operation of the relay, or other device, which energizes the light when aircraft electrical power is lost. This test shall be conducted while at the altitude and temperature conditions listed. At the end of the tests the light shall be removed from the test chamber and examined. There shall be no crazing, dimensional changes, or other defects. The light shall be returned to room temperature and checked for proper operation.

4.5.2.4 Humidity. The light shall be subjected to the humidity test in accordance with MIL-STD-810, method 507, procedure II, except that the test conditions shall be maintained only for a 72-hour period. The assembly shall not be adversely affected by the test. The lamp

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shall not be turned on during the soak period, but shall be turned on during the tests to check proper operation.

4.5.2.5 Sand and dust. The light shall be subjected to the sand and dust test in accordance with MIL-STD-810, method 510, procedures I and II, at 77 °F (25 °C), except that the test shall be of 4 hours duration. At the conclusion of the test, the assembly shall not be adversely affected by the test.

4.5.2.6 Salt spray. The light shall be subjected to the salt fog test in accordance with ASTM B117, for a period of 50 hours. At the end of the test, the assembly shall operate satisfactorily, and there shall be no evidence of excessive corrosion or failure of any part to function properly.

4.5.2.7 Explosion proof. The light shall be subjected to an explosive atmosphere test in accordance with MIL-STD-810, method 511, procedure I.

4.5.2.8 Vibration tests. The test specimen shall be mounted in normal operating position and subjected to vibration cycling in each of three mutually perpendicular planes. The cycling shall be from 10 to 500 cycles per second in 15 minute cycles for a period of 2 hours in each direction at an applied double amplitude of 0.036-inch or an applied acceleration of 10 G, whichever is the limiting value. Suitable instructions shall be provided to detect nuisance tripping of the contacts of the inertia switch and unwarranted opening or closing of the contacts of any relays that may be used in the light. No permanent deformation of any of the components of the light shall occur during this test.

4.5.2.9 Acceleration (functional test). The light shall be mounted on a centrifuge in normal operating position and subjected to acceleration in the horizontal plane. The speed of the centrifuge shall be gradually increased so that the operating point of the inertia switch may be determined. A reading shall be taken in each direction as the light is rotated in 45 degree increments around the axis of the inertia switch. During this test, the light shall not come on at any time before an acceleration of 1.5 G is applied and shall not require more than 2.5 G to become energized. The light shall remain on when subjected to an acceleration of 10 G.

4.5.2.10 Shock. The light shall be subjected to shock while mounted as it would normally be mounted while in service. The shock-testing machine shall be designed and fabricated in accordance with MIL-STD-202 or equivalent. The test unit shall be subjected to 18 impact shocks of 15 G acceleration, each shock impulse having a time duration of 11 ±1 milliseconds. The intensity shall be within ±10 percent when measured with a filter having a bandwidth of 5 to 100 cycles per second. The maximum acceleration shall be reached in approximately 5.5 milliseconds. The shock shall be applied in the following directions:

a. Vertically, 3 shocks in each direction.

b. Parallel to the major horizontal axis, 3 shocks in each direction.

c. Parallel to the minor horizontal axis, 3 shocks in each direction.

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The test unit shall not suffer damage or subsequently fail to provide the performance specified by the detail specification.

4.5.2.11 Underwater operation. When mounted on a suitable bracket and with new batteries installed, the light shall be turned on manually and then submerged to a depth of 3.5 feet in a tank containing a salt-water solution (3.5 percent sodium chloride by weight). While immersed, the light shall operate satisfactorily for a period of 10 minutes. For lights that are considered watertight, satisfactory operation may be construed as meaning that the light produces a visible beam and an inspection shows that no appreciable leakage occurred during the 10 minute period. For lights that are not considered watertight, the voltage at the lamp socket shall be measured continuously during the 10 minute period. If the test specimen provides a visible beam of light and the voltage does not fall below 2 volts during the immersion, the operation shall be considered satisfactory.

4.5.3 Illumination (photometric). The light shall be subjected to a photometric test to determine compliance with the light distribution specified in 3.5.1. The test shall be conducted with the lamp operated at rated voltage and with the light mounted in normal operating position. The candlepower distribution in a random plane passing through the axis of symmetry of the lamp shall be determined. The test shall then be repeated in order to obtain light distribution data in a plane that intersects the first data plane at an angle of 90 degrees and also passes through the axis of the lamp.

4.5.4 Portability. The light, in operating condition, shall be mounted on a bracket and oriented in the same manner as it would be over a normal exit in the aircraft. The device holding the light in its mounted position shall be checked to determine that the light can be easily and quickly removed. The disconnect on the terminal box shall operate properly. The light shall be operable after separation occurs.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The LEU-l/A light covered by this specification is intended for use in automatically illuminating the exits of cargo-type aircraft immediately after a crash landing.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of specification.
b. When first article testing is required (see 3.1).
c. When batteries are required (see 3.14).
d. Packaging requirements (see 5.1).

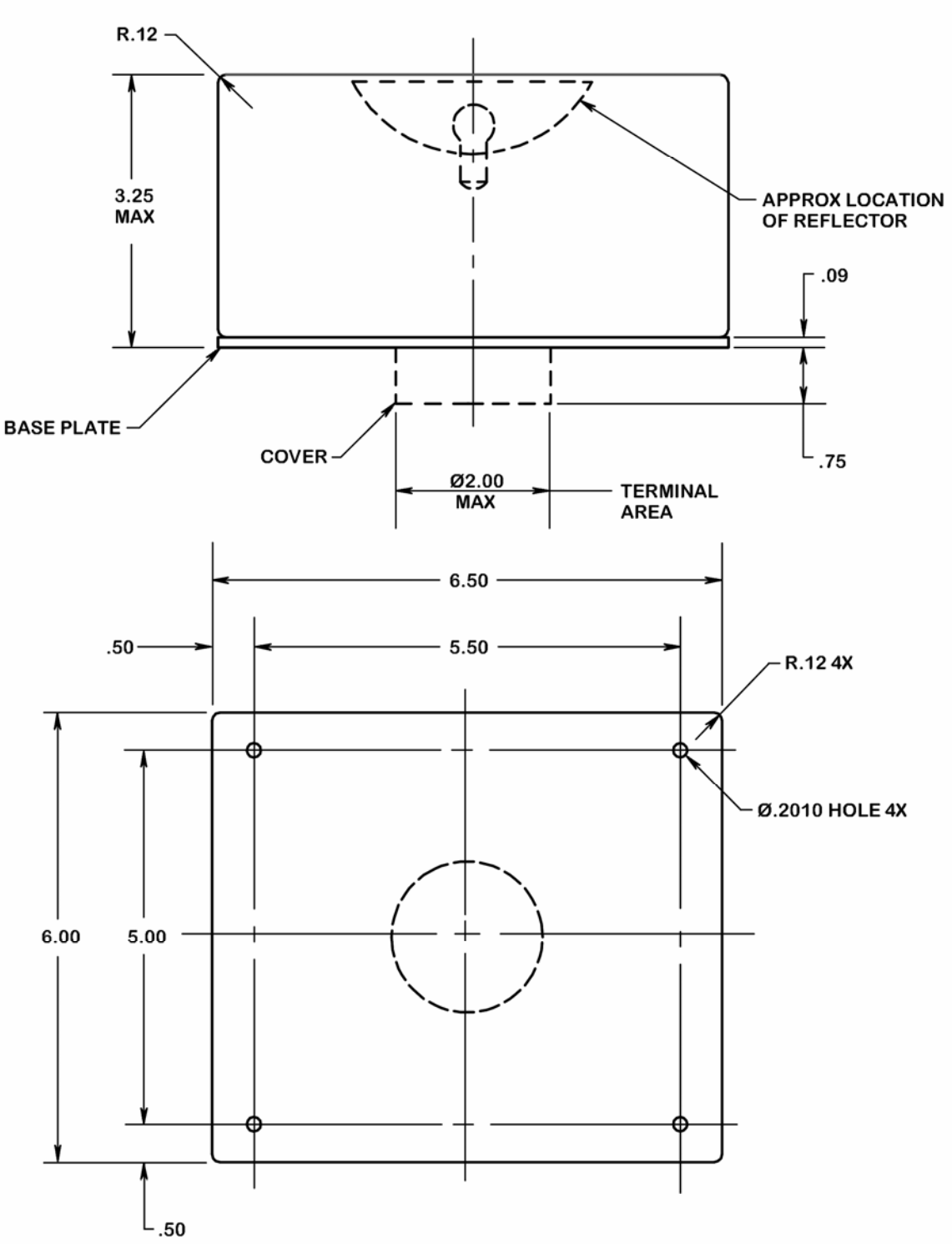
6.3 First article tests. Three lights will be required as first article samples to be subjected to the requirements of this specification. The point of inspection for these tests will be specified.

6.4 Subject term (key word) listing.

Switch, inertia

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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NOTES:
1. Dimensions are in inches.
2. Tolerances: Decimals ±.02.

FIGURE 1. Housing and base plate.

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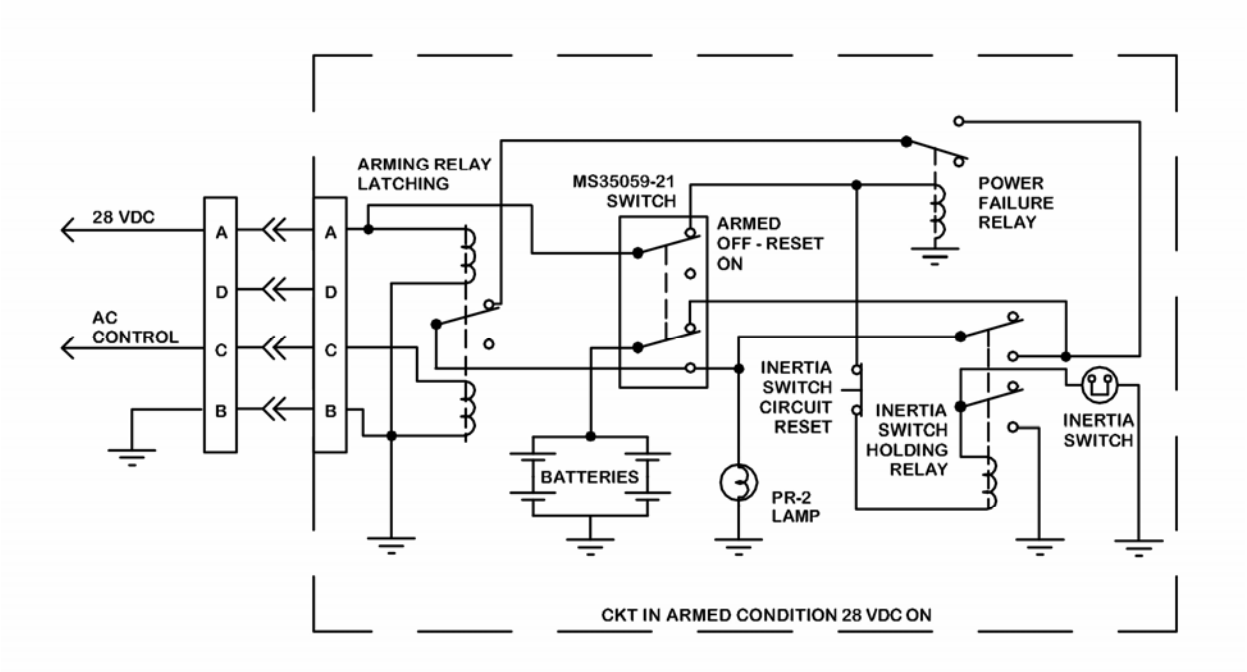


FIGURE 2. Schematic light circuit for type LEU-1/A.

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